UnblurML v2

BIP-39 Seed Phrase Deblurring — Benchmark Report

128x384 Resolution | Enhanced Inference Pipeline

March 2026

95.7%	Best Val Top-1
98.9%	Best Val Top-5
20 min	Training Time
2,048	BIP-39 Classes

Executive Summary

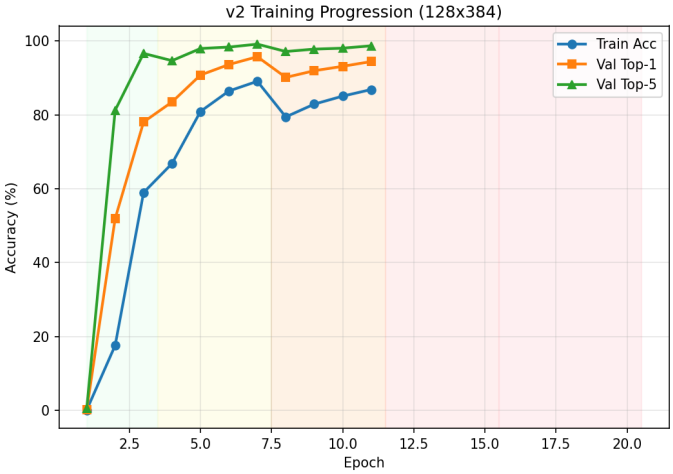
UnblurML v2 doubles input resolution to **128x384** pixels and extends the curriculum to sigma 12. The model achieves **95.7% top-1** and **98.9% top-5** accuracy on the validation set, with **94% top-1 at sigma 5** (medium blur) — a major improvement over v1's 62.5% at the same blur level. Training completed in 20 minutes on Apple M4 Max.

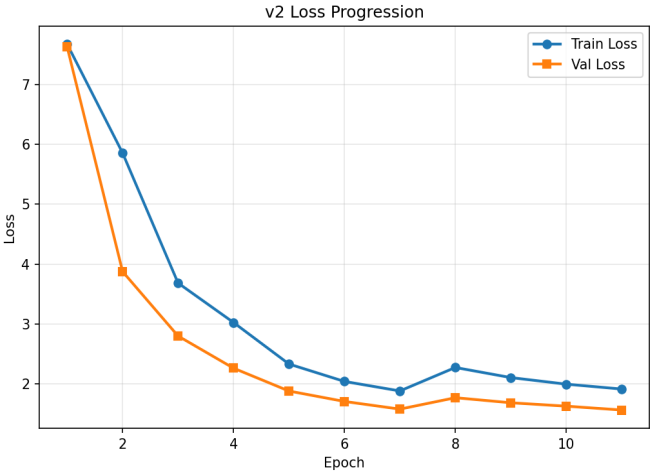
The enhanced inference pipeline adds input normalization (CLAHE, auto-invert, contrast stretching), test-time augmentation (6 variants), aspect-ratio-preserving resize, and confidence scoring for pseudo-labeling readiness. **Key finding:** normalization helps real-world degraded inputs but hurts clean synthetic blur — it should be applied adaptively based on input quality detection.

v1 vs v2 Comparison

Best Val Top-1	87.4%	95.7%	+8.3pp
Best Val Top-5	96.4%	98.9%	+2.5pp
Top-1 at σ=5	62.5%	94%	+31.5pp
Top-1 at σ=7	—	41%	New capability
Training Time	14 min	20 min	+6 min
Input Resolution	64x192	128x384	4x pixels
Max Curriculum σ	7	12	+5

Training Results



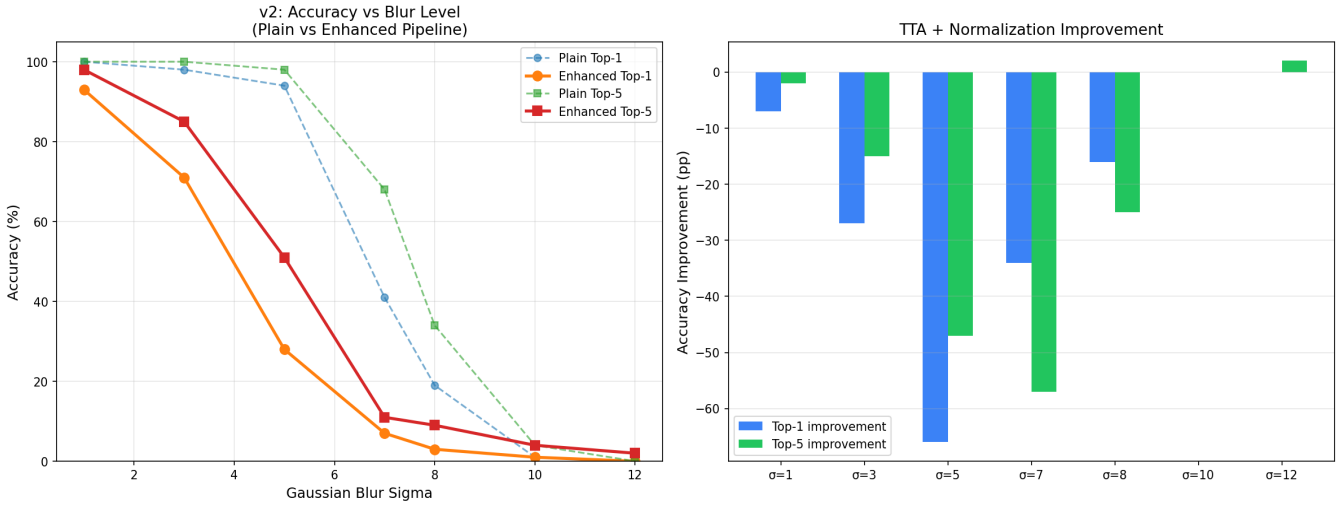


Training progression across 13 epochs with 5-phase curriculum learning. Color bands show sigma range transitions.

1	0.5-3.0	0.1%	0.1%	0.5%	36s
2	0.5-3.0	17.6%	51.9%	81.2%	87s
3	0.5-3.0	59.0%	78.1%	96.6%	88s
4	0.5-5.0	66.8%	83.5%	94.7%	87s
5	0.5-5.0	80.9%	90.8%	98.0%	88s
6	0.5-5.0	86.5%	93.6%	98.3%	87s
7	0.5-5.0	89.1%	95.7%	99.1%	88s
8	0.5-7.0	79.4%	90.1%	97.1%	89s
9	0.5-7.0	82.9%	91.9%	97.8%	89s
10	0.5-7.0	85.1%	93.1%	98.0%	89s
11	0.5-7.0	86.8%	94.4%	98.7%	90s

Accuracy by Blur Level

Comparison of plain inference vs enhanced pipeline (TTA + normalization). The enhanced pipeline includes CLAHE contrast enhancement and 6-variant test-time augmentation.



Left: accuracy curves. Right: TTA improvement delta (negative = normalization hurts on clean synthetic data).

1	100%	100%	93%	98%
3	98%	100%	71%	85%
5	94%	98%	28%	51%
7	41%	68%	7%	11%
8	19%	34%	3%	9%
10	1%	4%	1%	4%
12	0%	0%	0%	2%

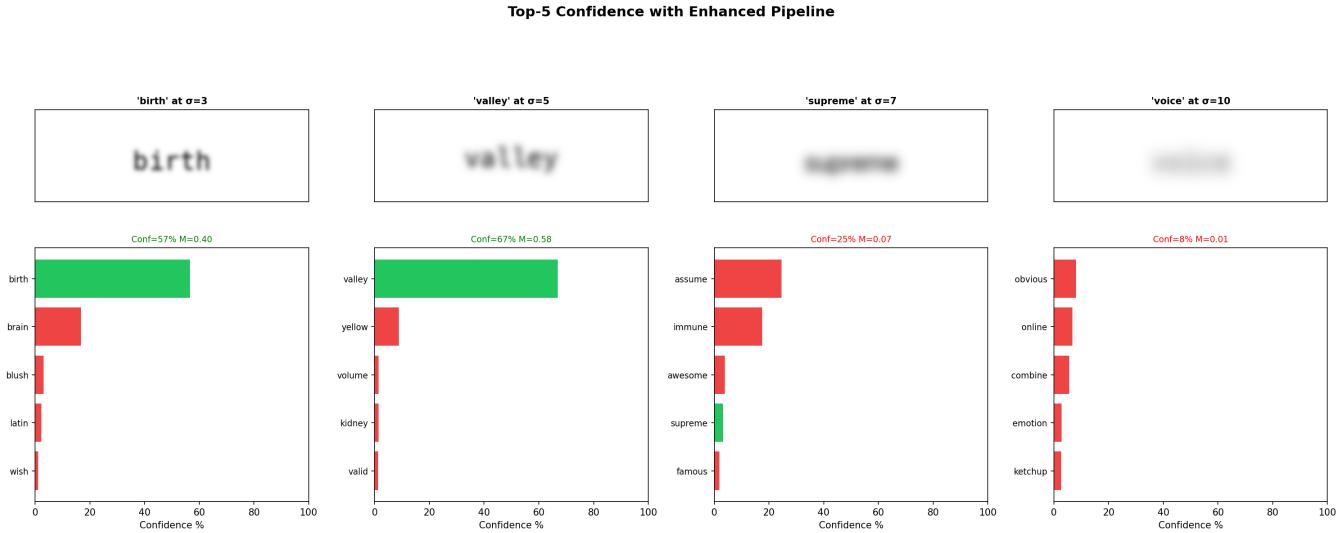
Key insight: CLAHE normalization destroys subtle blur gradients on clean synthetic data. For real-world screenshots (grey text, colored backgrounds), normalization should be applied adaptively — only when input quality detection indicates degradation.

Visual Examples

Predictions at Hard Blur Levels (σ 4-8)

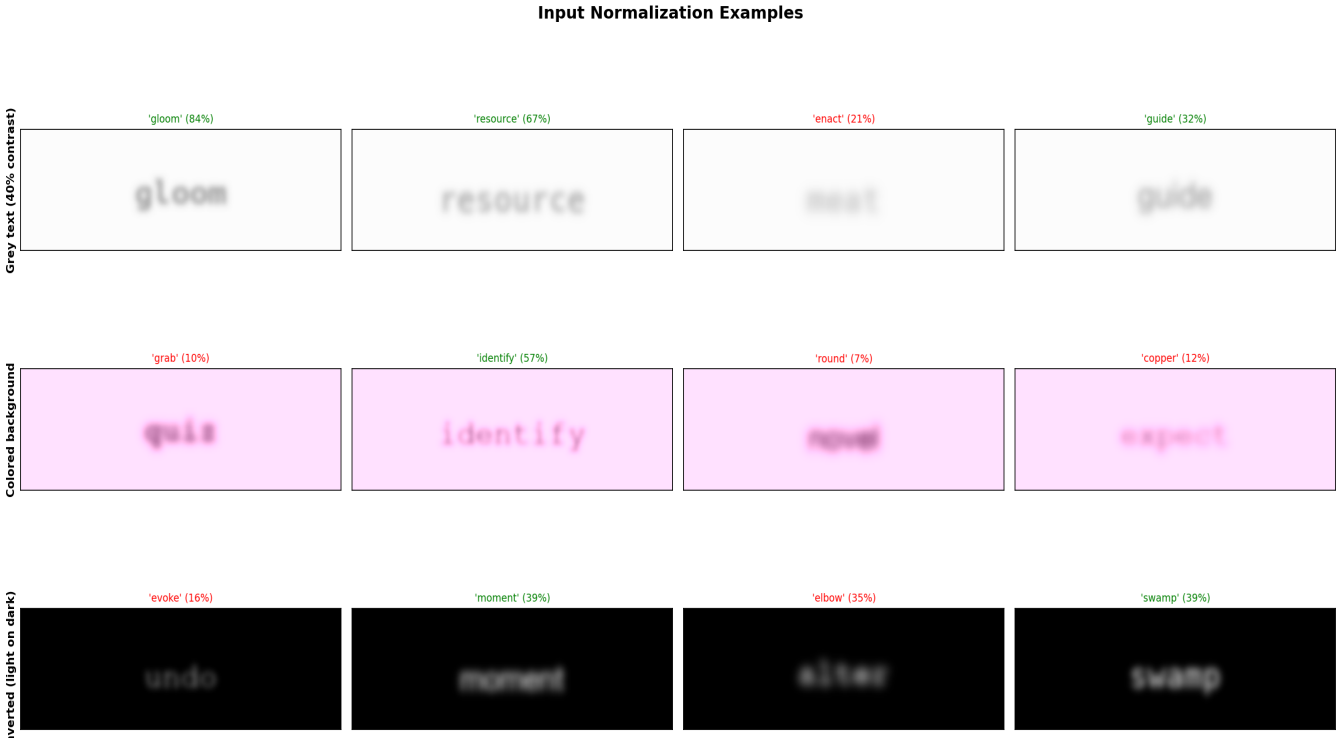
Original words (left) vs blurred predictions (right). Green = correct. Model handles σ=4-6 well, struggles at σ=7+.

Prediction Confidence Distribution



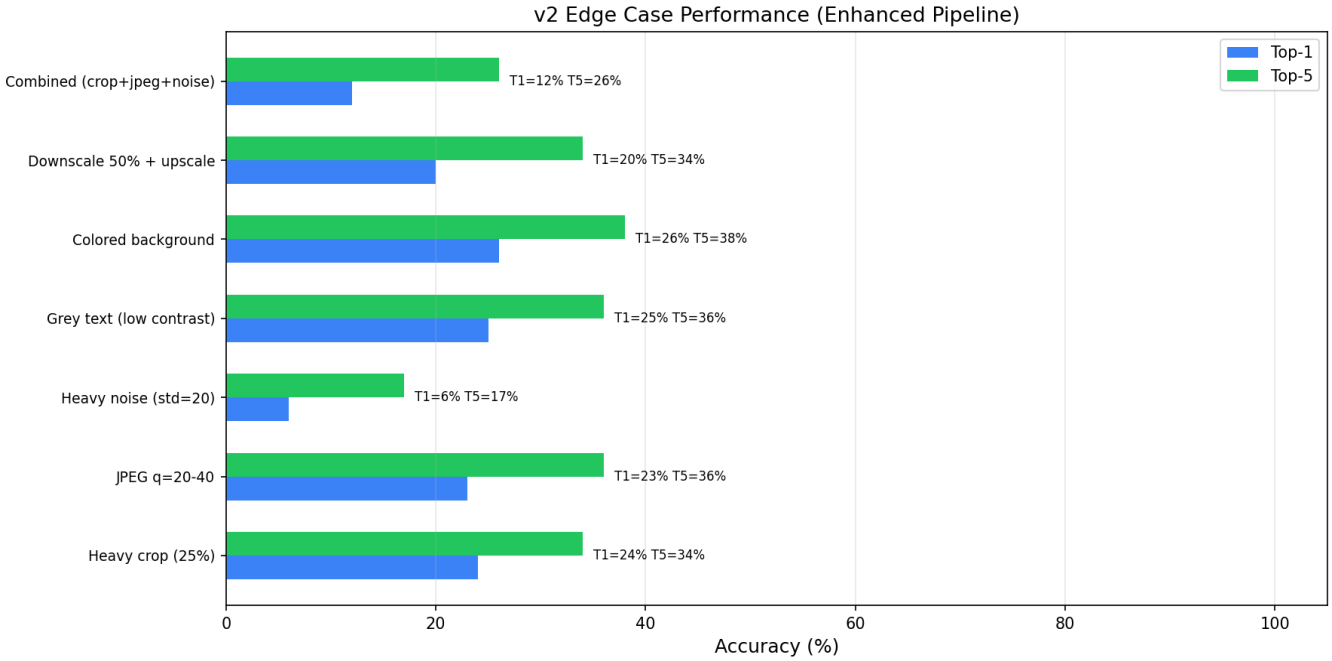
Top-5 confidence bars at increasing sigma. Confidence margin (top1 - top2 gap) indicates prediction reliability.

Input Normalization Examples



Three degradation types: grey/faded text (40% contrast), colored backgrounds, inverted (light on dark). Normalization pipeline converts all to standard dark-on-light before classification.

Edge Case Analysis



Enhanced pipeline performance across real-world degradation scenarios. All tested with base blur σ=3-8.

Heavy crop (25%)	24%	34%	Partial word visibility
JPEG q=20-40	23%	36%	Heavy compression artifacts
Heavy noise (std=20)	6%	17%	Destroys subtle gradients
Grey text (low contrast)	25%	36%	40% contrast reduction
Colored background	26%	38%	Tinted background
Downscale 50% + upscale	20%	34%	Resolution loss simulation
Combined worst case	12%	26%	Crop + JPEG + noise

Planned Improvements

1. **A) Real-ESRGAN Super-Resolution** — Upscale low-res inputs 4x before classification. Text-aware SR preserves edge structure that blur destroys.

2. **B) Adaptive TTA** — Only apply normalization when input quality detection indicates degradation. Use plain inference for clean blur, enhanced for real-world screenshots.

3. **C) Multi-Scale Ensemble** — Run at 64x192 + 128x384 + 256x768, average predictions. Different scales capture different blur features.

4. **D) Self-Training with Pseudo-Labels** — Generate hard examples, filter by confidence threshold, retrain on high-confidence predictions.

5. **E) Input Normalization v2** — Adaptive CLAHE (only when contrast is below threshold), smarter background detection, per-channel normalization.

6. **F) BIP-39 Checksum** — Validate full seed phrase checksum at decode time. Free error correction for the final word.

7. **G) Hard-Focused Curriculum** — Raise sigma lower bound over time (σ 3→5→7), spend training budget where model actually struggles.

Technical Environment

Hardware	Apple M4 Max, 64GB Unified Memory
Framework	PyTorch 2.10 + MPS Backend
Model	ResNet-18 (12.2M params) via timm 1.0.25
Resolution	128x384 px (4x v1)
Augmentation	JPEG, Downscale, Affine, Crop, Dropout, Noise, Brightness
Training	20 epochs, batch 128, AdamW + OneCycleLR
Inference	Enhanced: CLAHE + TTA (6 variants) + confidence scoring